

PORI, Finland

WMO: 029520

Lat: 61.467N

Lon: 21.788E

Elev: 13

StdP: 101.16

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.1	-17.9	-23.7	0.4	-21.1	-20.2	0.6	-17.5	10.5	0.8	9.4	0.4	1.8	130	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	26.8	18.3	24.9	17.4	23.0	16.4	19.6	24.3	18.5	22.9	17.5	21.7	4.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.0	21.5	16.8	12.0	20.7	15.8	11.2	19.5	56.0	24.1	52.2	22.7	49.1	21.4	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.0	7.1	-23.3	29.4	4.2	2.4	-26.3	31.1	-28.7	32.5	-31.1	33.9	-34.1	35.6		
(5)			-23.5	21.2	4.1	1.5	-26.5	22.2	-28.9	23.1	-31.2	23.9	-34.2	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-4.7	-5.0	-1.7	4.0	9.4	13.7	17.1	15.8	11.1	5.4	1.2	-2.1		
	DBStd	8.76	6.02	6.14	4.35	3.49	4.02	3.28	3.01	2.95	3.27	3.94	4.38	5.69		
	HDD10.0	2296	456	419	362	183	59	4	0	1	26	147	264	374		
	HDD18.3	4752	714	653	621	430	277	143	62	90	217	400	514	632		
	CDD10.0	624	0	0	0	3	41	117	219	179	59	6	0	0		
	CDD18.3	39	0	0	0	0	1	5	22	11	0	0	0	0		
	CDH23.3	347	0	0	0	0	18	54	190	84	1	0	0	0		
	CDH26.7	59	0	0	0	0	2	8	39	10	0	0	0	0		
	WSAvg	3.3	3.4	3.4	3.4	3.3	3.4	3.4	3.0	2.9	3.0	3.1	3.4	3.6		
	PrecAvg	612	50	36	33	29	34	55	67	73	55	72	57	53		
	PrecMax	755	83	78	58	68	87	112	124	155	123	164	142	103		
	PrecMin	515	7	1	5	1	4	17	2	30	11	21	9	6		
	PrecStd	61	21	21	13	16	21	26	33	34	35	31	29	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	5.8	10.1	19.2	25.9	27.8	30.1	28.0	21.9	15.1	10.8	8.1		
		MCWB	4.4	3.7	5.3	10.9	15.6	18.0	20.0	19.4	15.9	12.4	9.6	7.3		
	2%	DB	4.1	4.2	7.0	15.1	22.2	24.9	27.4	25.7	19.8	13.2	9.1	6.9		
		MCWB	3.2	2.9	4.0	8.9	13.6	16.6	18.9	18.0	15.0	11.5	8.2	6.2		
	5%	DB	3.1	3.1	5.1	12.8	19.4	22.2	25.2	23.7	18.0	12.1	8.0	5.8		
		MCWB	2.4	2.2	2.8	7.5	12.5	15.0	17.8	17.3	14.2	10.7	7.4	5.1		
	10%	DB	2.1	2.1	3.9	10.4	16.9	20.2	23.2	21.9	16.5	11.1	7.0	4.2		
		MCWB	1.6	1.3	2.0	6.3	11.3	14.1	17.0	16.6	13.6	9.7	6.4	3.6		
	0.4%	WB	4.9	4.0	5.9	11.6	17.2	19.6	21.5	20.9	16.8	13.0	9.9	7.4		
		MCDB	5.1	4.7	9.1	17.2	23.1	25.5	27.4	25.0	20.1	14.6	10.4	7.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	3.3	3.1	4.2	9.6	15.2	17.6	20.1	19.3	15.8	12.0	8.3	6.0		
		MCDB	3.7	3.8	6.2	14.3	20.6	22.7	25.1	23.5	18.6	13.1	8.8	6.4		
	5%	WB	2.4	2.2	3.1	8.0	13.2	16.2	18.9	18.2	14.9	11.1	7.3	4.7		
		MCDB	2.9	3.0	4.7	11.7	18.1	20.6	23.3	22.2	17.3	12.2	7.7	5.2		
	10%	WB	1.5	1.4	2.1	6.6	11.6	14.9	17.8	17.2	14.0	10.0	6.2	3.6		
		MCDB	2.0	1.9	3.4	9.8	16.0	19.0	22.3	20.8	16.1	10.8	6.7	4.1		
	5% DB	MDBR	5.5	5.9	7.2	8.8	9.8	9.0	9.1	9.2	8.1	5.9	4.2	4.9		
		MCDBR	4.6	5.3	8.2	13.2	13.6	13.0	12.7	12.3	10.1	6.5	4.2	4.3		
		MCWBR	4.3	4.5	5.7	7.6	6.9	6.3	5.6	5.9	6.0	5.1	3.9	4.0		
		MCWBR	4.3	4.9	7.0	11.8	11.7	10.9	10.7	10.3	8.4	5.9	4.1	4.0		
(40) Clear-Sky Solar Irradiance	taub		0.264	0.309	0.329	0.368	0.365	0.357	0.383	0.370	0.345	0.327	0.284	0.352		
		taud	2.136	2.235	2.317	2.327	2.410	2.463	2.413	2.440	2.496	2.395	2.191	2.530		
	Ebn at Noon		442	656	786	822	857	871	833	815	772	639	410	286		
		Edh at Noon	37	66	85	103	102	99	102	92	73	57	35	24		
(44) All-Sky Solar Radiation	RadAvg		0.20	0.75	2.27	3.87	5.41	5.89	5.45	4.25	2.52	1.04	0.27	0.12		
	RadStd		0.03	0.15	0.27	0.40	0.48	0.37	0.49	0.40	0.27	0.18	0.05	0.01		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (50 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page